

Week 10 — Activity: Stability, Time Stepping, and Information Transfer

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

To develop physical intuition for numerical stability in explicit finite difference schemes.

In a time-stepping model, temperature is updated step-by-step. Each step uses information from neighbouring nodes.

Think of heat as spreading gradually through the system.

Part A: Prediction

Consider two cases:

- very small time steps
- very large time steps

What do you expect to happen in each case?

Part B: Physical reasoning

Task: Explain why heat cannot move arbitrarily far in a single time step.

What determines how quickly heat diffuses through a material?

Part C: Stability condition

For a one-dimensional explicit scheme:

$$\Delta t < \frac{\Delta z^2}{2\kappa} \quad (1)$$

Tasks:

- Why does the allowable time step depend on Δz ?
- Why does it depend on thermal diffusivity κ ?

Part D: Interpretation

Task: Interpret the stability condition as a limit on how fast thermal information can move between nodes.

Why can heat not “jump” over nodes in a single time step?

Part E: Geological interpretation

Explain:

- Why are sedimentary basins (low κ) easier to model?
- Why do high-diffusivity materials require smaller time steps?

Key Insight

Numerical instability occurs when the model allows heat to propagate faster than physically possible.